1. why is Queue better than global variables

-Queues allow the RTOS to ensure that shared
data is transferred safely between tasks without
corruption or overwriting. They provide controlled
access to data so tasks can read messages without
the risk of other tasks modifying or consuming the
data simultaneously.

2. describe pros and cons of queue by copy vs
queue by reference

queue by copy

pros:
   -each queue gets its own copy of the data
   -no threat of queue data getting overwritten
cons:
   -overhead from copying data from large points of data
   -high memory

queue by reference
pros:
   -fast since only a pointer is copied
   -potentially memory efficient
cons:
   -more complex and error prone
   -data can be written over in the queue

3. In FreeRTOS, what is typically the scope of a queue
declaration? Why?

-it is declared as a global or a file variable. this allows
it to be shared between tasks while providing safe data
transfer between tasks.

4. What is the difference between the FreeRTOS API
functions xQueueSendToFront() and xQueueSendToBack()?

xQueueSendToBack()- this is standard use case. since most
queues are essentially FIFO's, this puts data in the back of
the line.

xQueueSendToFront()- this essentially line cuts the FIFO or
queue. if X is the front of queue and this function is called
for Y. then Y is first and X is second.

5. Describe a use case for using a queue as a mailbox.

- when you have a single point of data that changes value
and you dont need to record all values that it has been.
for instance a flag where you don't care about the sequence
of flag changes. Thus you can just care about the current value.